Signorini's Problem

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Abstract

This is the abstract.[1]

1 Introduction

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2 Mathematical Framework

2.1 Sobolev Spaces and the Trace Operator

In this subsection, we will make a brief excursion into selected topics in functional analysis, providing the necessary theory for the subsequent subsections. In classical analysis, derivatives are defined pointwise and require a certain level of smoothness. However, many functions of interest fall outside this framework. To extend the concept of differentiation to a broader class of functions, we can weaken what it means to *have a derivative*. The key idea is to characterize derivatives not by their pointwise behavior, but by how they act under integration against *test functions*. Using integration by parts, one can express the derivative of a smooth function in a form that remains meaningful even when the function itself is not differentiable in the classical sense. This leads naturally to the concept of weak derivatives, which allow us to treat integrable functions as *differentiable* in a generalized sense. We start by defining two important function spaces.

Definition 2.1 (Locally Integrable Functions)**.** *The space $L^1_{\mathrm{loc}}(\Omega)$ of all* `locally integrable` *functions on Ω is defined as*

$$L^1_{\mathrm{loc}}(\Omega) := \left\{ f : \Omega \to \mathbb{R} \ \text{measurable} \ \middle| \ f|_K \in L^1(K) \ \forall K \subset \Omega,\, K \ \text{compact} \right\}.$$

Definition 2.2. *The space $\mathcal{C}_c^\infty(\Omega)$ contains all infinitely often differentiable functions with compact support in Ω, i.e.*

$$\mathcal{C}_c^\infty(\Omega) = \{ f \in \mathcal{C}^\infty(\Omega) \mid \operatorname{supp}(f) \subset \Omega \ \text{compact} \}$$

and $u \in \mathcal{C}_c^\infty(\Omega)$ is called a `test function`*.*

With these functions in hand, we can now extend the classical concept of differentiability to the concept of *weak* differentiability.

Definition 2.3 (Weak Derivatives)**.** *Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^d$. A function $g \in L^1_{\mathrm{loc}}(\Omega)$ is called* `weak derivative` *of u of order α if*

$$\int_\Omega u \, \partial^\alpha \varphi \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega g \, \varphi \, \mathrm{d}x \quad \forall \varphi \in \mathcal{C}_c^\infty(\Omega)$$

If such g exists, it is unique up to null sets and we write $\partial^\alpha u = g$ and keep the notation the same as in the strong sense of derivatives.

Of course, there are also spaces of weakly differentiable functions, named after the Soviet mathematician Sergei Lvovich Sobolev (1908–1989).

Definition 2.4 (Sobolev Spaces)**.** *For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$ we define the* `Sobolev space` *$W^{k,p}(\Omega)$ to be the space of every L^p-function whose weak derivatives up to order k are also L^p-functions, i.e.*

$$W^{k,p}(\Omega) := \{ u \in L^p(\Omega) \mid \partial^\alpha u \in L^p(\Omega) \ \forall |\alpha| \le k \}$$

and we equip it with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \le k} \|\partial^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p}, \quad 1 \le p < \infty$$

and

$$\|u\|_{W^{k,\infty}(\Omega)} := \max_{|\alpha| \le k} \|\partial^\alpha u\|_{L^\infty(\Omega)}.$$

The special case $p = 2$ provides us with a Hilbert space $H^k(\Omega) := W^{k,2}(\Omega)$ together with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \leq k} \langle \partial^\alpha u,\, \partial^\alpha v \rangle_{L^2(\Omega)}$$

For the following purpose, we will have to consider the the space

$$H^1(\Omega) = \left\{ u \in L^2(\Omega) \;\middle|\; \nabla u \in L^2(\Omega; \mathbb{R}^d) \right\}$$

together with the induced norm

$$\|u\|_{H^1(\Omega)} = \left(\|u\|^2_{L^2(\Omega)} + \|\nabla u\|_{L^2(\Omega; R^d)} \right)^{1/2}$$

where the second L^2–term is understood as

$$\|\nabla u\|_{L^2(\Omega; \mathbb{R}^d)} = \left(\int_\Omega |\nabla u(x)|^2 \, \mathrm{d}x \right)^{1/2}.$$

3 Signorini's Problem

3.1 Physical and Geometric Setup of the Problem

Throughout the whole report, $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ denotes an open, bounded and connected *Lipschitz* domain. The later means, that the boundary $\Gamma := \partial\Omega$ can locally be described as the graph of a Lipschitz continuous function. This ensures for example, that outward normal vectors $\mathrm{n}(x) : \Gamma \to \mathbb{R}^d$ are well–defined almost everywhere. The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \,\dot{\cup}\, \overline{\Gamma_N} \,\dot{\cup}\, \overline{\Gamma_C}$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions. On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D,

$$\forall x \in \Gamma_D : \; x + u(x) = x \quad \Longrightarrow \quad \mathrm{Tr}(u)|_{\Gamma_D} = 0.$$

The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. We assume that Γ_C has positive $(d-1)$–Lebesgue measure. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\mathrm{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral* contact through *nonpenetration* and *complementarity*.

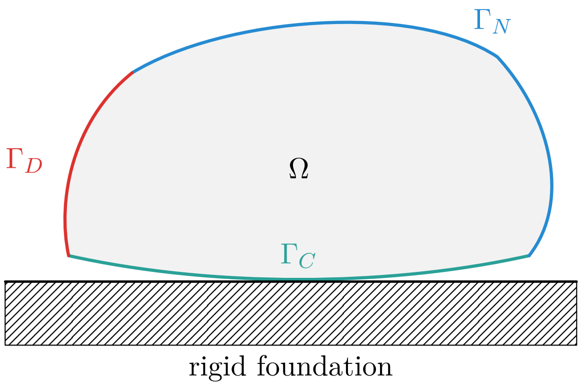

Figure 1: 2D Setup for Signorini's Problem.

3.2 Strong Formulation

3.3 Weak Formulation

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4 Outlook

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

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